PROJECT DESIGN PHASE OF LEAVE MANAGEMENT SYSTEM

System Architecture Design

The Leave Management System follows a layered architecture where employees submit leave requests through a custom form, requests are stored in a custom table, and approvals are managed through Flow Designer. Business Rules automate leave duration calculation, while Client Scripts perform form validation and ensure data accuracy.

The architecture is designed to provide scalability, maintainability, and efficient workflow automation while reducing manual effort.

System Architecture Diagram

Employee

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Leave Request Form

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Leave Request Table

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Client Script

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Business Rule

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Flow Designer

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Manager Approval

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Status Update

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Dashboard & Reports

Architecture Components

Employee

Submits leave requests through the ServiceNow application.

Leave Request Form

Provides a user-friendly interface for leave request submission.

Leave Request Table

Stores employee leave request information and approval details.

Client Script

Validates leave dates and mandatory fields before submission.

Business Rule

Calculates total leave days automatically based on selected dates.

Flow Designer

Automates approval workflows and status updates.

Dashboard

Displays leave request statistics and management reports.

Notifications

Sends automated notifications for submission, approval, and rejection events.

Database Design

A custom table was designed to store all employee leave request information.

Table Name

Leave Request

Table Fields

Request Number (Primary Key)

Employee Name

Employee ID

Leave Type

Start Date

End Date

Total Days

Reason

Approval Status

Created Date

Field Name	Data Type	Description
Employee Name	String	Employee name
Employee ID	String	Employee identifier
Leave Type	Choice	Type of leave
Start Date	Date	Leave start date
End Date	Date	Leave end date
Total Days	Integer	Calculated leave days
Reason	String	Leave reason
Approval Status	Choice	Approval state
Created Date	Date/Time	Request creation date

Form Design

A structured form was designed to collect leave request information from employees.

Form Layout

Section 1: Employee Details

• Employee Name

• Employee ID

Section 2: Leave Information

• Leave Type

• Start Date

• End Date

• Total Days

- Reason

Form Design Structure

Leave Request Form

Employee Details

- Employee Name
- Employee ID

Leave Details

- Leave Type
- Start Date
- End Date
- Total Days
- Reason

Submit

Design Considerations

- Easy navigation
- Mandatory field validation
- Reduced user errors
- User-friendly interface
- Responsive form layout
- Improved data accuracy

Application Design

A custom ServiceNow application was designed to provide self-service leave request management.

Application Name

Leave Management System

Form Fields

Employee Name – Single Line Text

Employee ID – Single Line Text

Leave Type – Choice

Start Date – Date

End Date – Date

Total Days – Integer

Reason – Multi Line Text

Approval Status – Choice

Application Modules

• Leave Request Creation

• My Leave Requests

• Approval Management

• Reports and Dashboard

Workflow Design

The workflow was designed using ServiceNow Flow Designer to automate the approval process.

Workflow Diagram

Leave Request Created

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Set Status = Pending Approval

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Ask For Approval

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If Approved

Yes → Update Status = Approved

No → Update Status = Rejected

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Send Notification

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Update Dashboard

Workflow Components

Trigger

Leave Request Created

Approval Action

Routes leave request to the reporting manager for approval.

Business Rule

Calculates total leave days automatically.

Client Script

Validates start date and end date.

IF Condition

Checks approval status.

Status Update

Updates request status after approval decision.

Notification Action

Sends approval or rejection notifications to employees.

Dashboard Reporting

Displays total, approved, rejected, and pending requests.

11. Design Advantages

Scalability

Supports increasing leave request volume without affecting performance.

Maintainability

Components can be modified independently with minimal impact on other modules.

Automation

Reduces manual effort through workflows, business rules, and automated notifications.

User Experience

Provides a simple, intuitive, and easy-to-use interface.

Transparency

Provides complete visibility into leave request status and approval progress.

Accuracy

Automatic leave calculation reduces errors and improves consistency.

Security

Ensures that only authorized users can access and manage leave request information.

Flexibility

Allows future enhancements and integration with additional HR processes.

Conclusion

The Project Design Phase defined the complete structure of the Leave Management System. The architecture, database design, form design, workflow design, validation logic, notification mechanism, and reporting dashboard provide a strong foundation for implementation.

The design ensures efficient leave request handling, approval automation, accurate leave calculation, secure data management, and real-time request tracking while maintaining scalability, reliability, and usability.